

5.2 Circle relationships

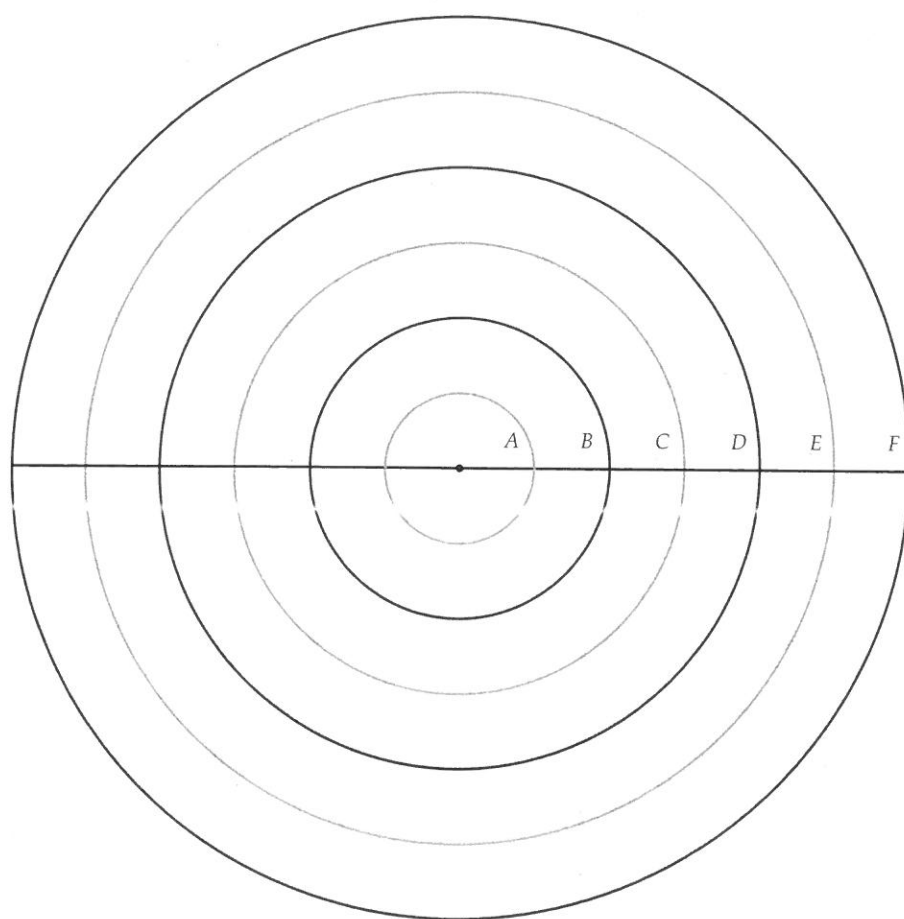
Answers
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Equipment required: Ruler, string, compass (optional), circular objects such as cups, plates, lids, drink bottles, wheels (optional) for Question 1; a scientific calculator for Question 7(b)

Fluency

- 1 Use your ruler to measure the diameter and radius of circles A–F given on the following page. Record your measurements in the table underneath. Measure and record the circumference of each circle by carefully laying the string along it. (A classmate can help you do this reasonably accurately.) Mark the beginning and end of the circumference on the string, then hold it straight against a ruler and read off the length of the circumference.

Alternatively, choose six circular objects, or use your compass to draw six circles of different sizes, then measure and record the diameter, radius and circumference using your ruler and string. (If you are using a compass, remember that the distance to which you open your compass is the same as the radius of the circle.)



Circle	Radius r (cm)	Diameter d (cm)	Circumference C (cm)
A			
B			
C			
D			
E			
F			

Hint

Can't remember what 'unit ratio' means? Turn to Section 4.3.

Understanding

- 2 (a) Use your completed table from Question 1 to describe any patterns or connections you can see between the numbers in the 'radius', 'diameter' and 'circumference' columns.
- (b) To see the connections more clearly, add two columns to the end of your table, with the headings $\frac{C}{d}$ and $\frac{C}{r}$. Calculate these two unit ratios for each circle that you measured, round off to two decimal places, and record your results in the new columns. These unit ratios give us a scale factor for the length of the circumference compared to the diameter or radius.
- 3 Use your completed table from Question 1 and the unit ratios you calculated in Question 2 to copy and complete the following sentences.
- (a) The circumference of a circle is approximately _____ times the size of its diameter.
- (b) The circumference of a circle is approximately _____ times the size of its radius.
- (c) The length of the diameter of a circle is _____ the length of its radius.

Reasoning

- 4 Based on your answers to Questions 2 and 3, if a circle has a diameter of 4 m, predict the lengths of the:
- (a) radius (b) circumference.
- 5 Based on your answers to Questions 2 and 3, predict what will happen to:
- (a) the diameter (b) the circumference
- when the radius of the circle is doubled.
- 6 (a) Explain whether or not the measurements you obtained for the circumference by measuring with string and a ruler are accurate.
- (b) How would your calculations of $\frac{C}{d}$ and $\frac{C}{r}$ be affected if you measured the circumference as being:
- (i) smaller than actual size (ii) bigger than actual size?
- 7 (a) The exact value of the ratio $\frac{\text{circumference}}{\text{diameter}}$ is represented by the symbol π (the Greek letter pi).
- π is an irrational number, which means that if written as a decimal, it is a non-terminating, non-recurring decimal. Describe what the digits in a 'non-terminating, non-recurring' decimal look like.
- (b) Find and press the ' π ' key on your calculator. Write the value of π as a decimal approximation, rounded to:
- (i) two decimal places (ii) three decimal places (iii) four decimal places.

Open-ended

- 8 (a) Choose a length for the diameter of a circle that lies in between the lengths of two diameters listed in the table you drew in Question 1. (For example, if your table contains diameters of 4 cm and 6 cm, you could choose 5 cm or 5.5 cm.) Use your circumference measurements for the two known diameters to estimate the circumference of a circle with your chosen diameter.
- (b) Repeat part (a) for a different value that lies in between two different diameters.